The Zeta Basis: a Family of Zeta Functions and Their Inner Product

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Abstract

The completed zeta function ξ is the $n = 0$ member of a family: summing the n-th Hermite function over the lattice and taking Mellin transforms yields $\Lambda_n(s) = \zeta(s)\,\mathcal{M}\psi_n(s)$, one "zeta function" for each even n, each satisfying a functional equation with eigen-sign $(-1)^{n/2}$. The extra zeros of Λ_n beyond the Riemann zeros are the zeros of a Meixner–Pollaczek polynomial and lie *provably* on the critical line (the local Riemann Hypothesis of Bump–Ng). The family is orthogonal: under the Mellin–Plancherel pairing renormalised by $|\zeta|^{-2}$ on the critical line — equivalently, under the Meixner–Pollaczek weight $|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|^2$ — one has $\langle \Lambda_m, \Lambda_n \rangle = \frac{1}{2}\delta_{mn}$, and the family is a basis of the even part of a Hilbert space $\mathcal{H}_\zeta$ of "zeta-like" functions. We record the definitions, the orthogonality, and the coordinates of ξ itself in the basis: geometric, with ratio $\sqrt{(2\pi-1)/(2\pi+1)}$.

1 The family

Let $\psi_n(x) = c_n H_n(x) e^{-x^2/2}$, $c_n = (2^n n!\sqrt{\pi})^{-1/2}$, be the orthonormal Hermite functions, and for even n set

$$\theta_n(x) \;=\; \sum_{m\in\mathbb{Z}} \psi_n(mx), \qquad x > 0.$$

For $n = 0$ this is the classical theta function up to the dilation $e^{-x^2/2}$ vs. $e^{-\pi x^2}$.

Definition 1.1. *For even n, the n-th zeta function is*

$$\Lambda_n(s) \;=\; \int_0^\infty \frac{\theta_n(x) - \psi_n(0)}{2}\, x^{s-1}\, dx \;=\; \zeta(s)\,\mathcal{M}\psi_n(s), \qquad \mathcal{M}\psi_n(s) = \int_0^\infty \psi_n(x)\, x^{s-1} dx, \quad (1)$$

the factorisation holding for $\operatorname{Re} s > 1$ by summing $\int_0^\infty \psi_n(mx)\, x^{s-1} dx = m^{-s}\mathcal{M}\psi_n(s)$, and elsewhere by continuation.

Λ_n is meromorphic with (at most) simple poles at $s = 1$ (from ζ) and $s = 0$ (from the Γ-factor below); it is regular and rapidly decaying on the critical line, which is all the inner product will see. Poisson summation and $\widehat{\psi_n} = (-i)^n \psi_n$ give $\theta_n(1/x) = (-1)^{n/2}\, x\, \theta_n(x)$ and hence the functional equation

$$\Lambda_n(1-s) \;=\; (-1)^{n/2}\, \Lambda_n(s)\,:$$

the family diagonalises the reflection $s \mapsto 1 - s$ exactly as the Hermite functions diagonalise Fourier. (Odd n give $\theta_n \equiv 0$; the odd half of the story belongs to $L(s, \chi)$ with $\chi(-1) = -1$, via the sign character at the archimedean place.)

2 The polynomial part and the local Riemann Hypothesis

From the companion paper on log-Hermite functions, for even n,

$$\mathcal{M}\psi_n(s) \;=\; c_n\, 2^{s/2-1}\,\Gamma\!\left(\tfrac{s}{2}\right) P_n(s), \tag{2}$$

with P_n a polynomial of degree $n/2$ in $s(1-s)$; e.g. $P_0 = 1$, $P_2 = 2(2s-1)$, $P_4 = 4(4s^2-4s+3)$, $P_6 = 8(2s-1)(4s^2-4s+15)$. Thus

$$\text{zeros of } \Lambda_n \;=\; \text{Riemann zeros} \;\cup\; \text{zeros of } P_n.$$

Bump–Ng proved that all zeros of P_n lie on $\operatorname{Re} s = \frac{1}{2}$ (the *local Riemann Hypothesis*). So every member of the family satisfies RH if and only if ζ does: the basis enlarges the zero set only by zeros that are certified.

On the critical line the P_n are classical: up to normalisation,

$$P_{2m}\!\left(\tfrac{1}{2}+i\tau\right) \;\propto\; P_m^{(1/4)}\!\left(\tfrac{\tau}{2};\,\tfrac{\pi}{2}\right), \tag{3}$$

the Meixner–Pollaczek polynomials, orthogonal on $\mathbb{R}$ with respect to $|\Gamma(\frac{1}{4}+ix)|^2\,dx$. (Check for $m=2$: $P_2^{(1/4)}(x;\frac{\pi}{2}) = \frac{2}{3}(1-8x^2)$, zeros $x = \pm\frac{1}{2\sqrt{2}}$, i.e. $\tau = \pm\frac{1}{\sqrt{2}}$ — the zeros of P_4.) The three-term recurrence and interlacing used by Bump–Ng are the Jacobi matrix of this orthogonal family.

3 The inner product

Definition 3.1. *Let L^2_ζ be the space of functions $F = \zeta \cdot G$ on the critical line with $G \in L^2\!\left(\frac{1}{2}+i\mathbb{R},\,\frac{d\tau}{2\pi}\right)$, with*

$$\langle F_1, F_2\rangle_\zeta \;=\; \frac{1}{2\pi}\int_{\mathbb{R}} \frac{F_1\!\left(\frac{1}{2}+i\tau\right)\overline{F_2\!\left(\frac{1}{2}+i\tau\right)}}{\left|\zeta\!\left(\frac{1}{2}+i\tau\right)\right|^2}\,d\tau. \tag{4}$$

The weight $|\zeta|^{-2}$ blows up at the Riemann zeros, but every member of L^2_ζ carries ζ as a factor, so the integrand is finite: the inner product simply strips the common arithmetic factor and applies Mellin–Plancherel to what remains.

Proposition 3.2. *$\langle \Lambda_m, \Lambda_n\rangle_\zeta = \frac{1}{2}\,\delta_{mn}$ for even m, n, and $\{\sqrt{2}\,\Lambda_n : n \text{ even}\}$ is an orthonormal basis of the even part of L^2_ζ (eigen-sign + functions under $s \mapsto 1-s$ generated by Mellin transforms).*

Proof sketch. By (1), $\Lambda_n/\zeta = \mathcal{M}\psi_n$, and Mellin–Plancherel (Fourier–Plancherel read through $x = e^t$) gives

$$\langle \Lambda_m, \Lambda_n\rangle_\zeta = \frac{1}{2\pi}\int_{\mathbb{R}} \mathcal{M}\psi_m\overline{\mathcal{M}\psi_n}\,d\tau = \int_0^\infty \psi_m(x)\,\psi_n(x)\,dx = \tfrac{1}{2}\,\delta_{mn},$$

the last step because $\psi_m\psi_n$ is even and $\{\psi_n\}$ is orthonormal on $\mathbb{R}$. Completeness: the even Hermite functions restricted to $\mathbb{R}_+$ span $L^2(\mathbb{R}_+)$, since restriction is an isomorphism from the even subspace of $L^2(\mathbb{R})$. □

Remark 3.3. Combining (2) and (3), the orthogonality can be written without ζ at all:

$$\langle \Lambda_{2j}, \Lambda_{2k} \rangle_\zeta \;\propto\; \int_{\mathbb{R}} P_j^{(1/4)}\big(\tfrac{\tau}{2}\big)\, P_k^{(1/4)}\big(\tfrac{\tau}{2}\big)\, \big|\Gamma\big(\tfrac{1}{4}+\tfrac{i\tau}{2}\big)\big|^2\, d\tau.$$

The critical line thus carries two structures at once: it is the Mellin–Plancherel *unitarity* line, and the *orthogonality* line of the polynomial parts of the basis.

4 Coordinates of ξ in the basis

Modulo the pole-killing prefactor $\frac{s(s-1)}{2}$, the completed zeta function is $\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s) = 2\,\zeta(s)\,\mathcal{M}g(s)$ with $g(x) = e^{-\pi x^2}$: a *dilated* Gaussian, not the basis Gaussian ψ_0. Its coordinates follow from the classical expansion of a Gaussian of width a in Hermite functions,

$$\big\langle e^{-ax^2/2},\, \psi_{2m} \big\rangle \;=\; \pi^{1/4}\sqrt{\frac{2}{1+a}}\;\frac{\sqrt{(2m)!}}{2^m m!}\,\Big(\frac{a-1}{a+1}\Big)^m,$$

with $a = 2\pi$:

$$\pi^{-s/2}\Gamma\big(\tfrac{s}{2}\big)\zeta(s) \;=\; \sum_{m\geq 0} g_{2m}\,\Lambda_{2m}(s), \qquad g_{2m} \;\propto\; \frac{\sqrt{(2m)!}}{2^m m!}\Big(\frac{2\pi-1}{2\pi+1}\Big)^m. \tag{5}$$

The coordinates of zeta in the zeta basis are geometric with ratio $q = \frac{2\pi-1}{2\pi+1} = 0.7254\ldots$ per index pair, i.e. $\sqrt{q} = 0.8517$ per Hermite index — the arithmetic object is a small, explicit perturbation of the $n = 0$ basis vector.

5 Outlook

Remark 5.1. For any even Schwartz $f = \sum_n a_n\psi_n$, linearity gives $\Lambda_f = \zeta\cdot\mathcal{M}f = \sum_n a_n\Lambda_n$, with $\|\Lambda_f\|_\zeta^2 = \frac{1}{2}\sum |a_n|^2$. The question "which Λ_f have all zeros on the line" splits into the Riemann zeros (common to all members) plus the zeros of $\sum_n a_n\,\mathcal{M}\psi_n$: which linear combinations of Meixner–Pollaczek polynomials (weighted by the common Γ-factor) are real-rooted — a question about orthogonal polynomials, where interlacing theory has teeth. This sharpens the open thread of the log-Hermite paper.

Remark 5.2. The truncation experiments of the companion papers can now be re-run *inside* the basis: expanding a target in $\{\Lambda_n\}$, every basis element already satisfies RH, and the balance-law analysis of spurious zeros applies verbatim to the polynomial parts $\sum_{n\leq N} a_n P_n$ on the critical line.

6 Numerical appendix: the approximation matrix of the family

Let Z_{nk} be the coefficients of the best degree-60 polynomial approximation of $F_n(t) = \xi_n(\frac{1}{2}+i\lambda t)$, $\lambda = 10$, in $L^2(\mathbb{R}, e^{-t^2}dt)$ (the log-Hermite polynomial approximation of the companion paper, applied to each family member). Since $\xi_n = (c_n/c_0)P_n\cdot\xi$ and multiplication by τ acts on Hermite coefficients as the tridiagonal position operator $(Tv)_j = \sqrt{j/2}\,v_{j-1} + \sqrt{(j+1)/2}\,v_{j+1}$, the whole matrix is generated from the single Ξ column β of the companion paper:

$$Z_n \;=\; \tfrac{c_n}{c_0}\,P_n\big(\tfrac{1}{2}+i\lambda T\big)\,\beta.$$

The arithmetic lives in one column; the rest is oscillator representation theory. Row n is supported on columns $k \equiv \frac{n}{2} \pmod 2$ (the functional-equation eigen-sign as column parity; entries '$\cdot$' below vanish identically).

$\log_{10}\lvert Z_{nk}\rvert$	k=0	6	12	18	24	30	36	42	48	54	60
$n=0$	−0.4	−1.1	−1.7	−2.2	−2.8	−3.4	−4.0	−4.6	−5.3	−6.0	−6.6
$n=2$	0.0	−0.2	−0.6	−1.1	−1.6	−2.1	−2.6	−3.2	−3.8	−4.5	$\cdot$
$n=4$	0.6	0.1	0.0	−0.2	−0.6	−1.1	−1.5	−2.1	−2.6	−3.2	−3.8
$n=6$	0.9	0.0	0.2	0.2	−0.1	−0.4	−0.8	−1.3	−1.8	−2.4	$\cdot$

The zeros of each row's approximant split exactly as the family's zeros do: the certified P_n zeros (captured to 3–4 digits, deep inside the capture zone), the Riemann zeros (captured up to the balance height $T_c \approx 17$), and truncation artifacts beyond. Truncation treats provable and conjectural zeros identically — it sees only the balance law.

n	certified zeros of P_n	captured	first Riemann zeros captured
0	—	—	14.1348, 21.029
2	0	0.0000	14.1348, 20.964
4	$1/\sqrt{2} = 0.7071$	0.7069	14.1332, 21.061
6	0, $\sqrt{7/2} = 1.8708$	0.0000, 1.8717	14.1330, 20.943

(True values: 14.1347, 21.0220; the third Riemann zero 25.0109 lies past the crossover and is not captured by any row at degree 60.) The rows decay more slowly as n grows — the $(\lambda\sqrt{2k})^{n/2}$ amplification of the Jacobi action — so the capture of the second Riemann zero degrades visibly down the column. Code: `family_matrix.py`.